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Pharmacist–physician collaborative care model for patients with uncontrolled type 2 diabetes in Brazil: results from a randomized controlled trial

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Abstract

Rationale, aims and objectives This study aimed to evaluate the effect of a pharmacist–physician collaborative care model on clinical outcomes in patients with uncontrolled type 2 diabetes and determine characteristics that influence this effect.

Methods A randomized controlled trial was conducted in a secondary care clinic for 80 patients with type 2 diabetes, aged 40–79 years and glycosylated haemoglobin (A1C) level $\geq$ 7.0%. The intervention group received individual, face-to-face pharmaceutical consultations and remote telephone support after a routine visit. The main measures were clinical outcomes (A1C, blood pressure, LDL cholesterol) and process indicators (medication adherence, medication regimen complexity, use of medicines). Multiple regression models were used to determine the variables that could explain the reduction and individualized control of A1C.

Results From the initial sample of 80 patients, 73 completed this study. Compared with usual care, patients in the intervention group showed greater reduction in A1C (−0.79 vs. −0.16; $P = 0.010$); and an increase in the percentage of patients achieving the individualized goal of A1C (25.0% vs. 5.4%; $P = 0.020$). In addition, there was an increase in the percentage of adherent patients and in the average scores of medication adherence. Participation in the intervention group, higher baseline A1C levels and greater change in medication adherence were all significant predictors of improvement in A1C levels.

Conclusions The results suggest that the collaborative care model proposed is feasible and more effective than the usual care in the reduction and individualized control of A1C levels in patients with uncontrolled type 2 diabetes.

Introduction

Type 2 diabetes is associated with increased morbidity and mortality worldwide through direct clinical sequelae and increased mortality from cardiovascular and kidney diseases [1]. Recent estimates confirm the large and increasing global health burden of diabetes, and it is likely to continue to grow substantially in the coming decades, especially in developing countries. In Brazil, it is projected that the number of people with this chronic illness will increase by 61% by 2035 [2]. Therefore, health systems will have to develop strategies to improve diabetes detection and management to slow progression of long-term complications.

Despite data supporting the strict control of diabetes and increased awareness of the benefits of improved metabolic control, only 26% of patients with type 2 diabetes in Brazil have achieved the recommended goal of a glycosylated haemoglobin (A1C) levels < 7% [3]. Unsatisfactory outcomes could reflect the failure of diabetes self-management as well as inadequate intervention strategies by health-care providers [4]. Growing evidence suggests multidisciplinary team-based care models (i.e. including a member other than the physician, such as a pharmacist, where all professionals have an active participation in management of patients) are the most effective quality strategies to meet the health complex needs of patients with type 2 diabetes and improve their outcomes [5].

Pharmacists have expanded their services further than drug dispensing, and they often participate directly in the management of patients with the delivery of clinical services [6]. In Brazil,

however, the standard pharmaceutical services are drug centred; they include primarily activities related to the acquisition and inventory control of drugs [7]. Therefore, clinical services have not been established as a common practice among Brazilian pharmacists [8]. Facing this challenge, an opportunity has emerged for this professional to have a more important role in the healthcare team, so that the value of the clinical services can be confirmed and professional competence established in the management of patients with diabetes.

Randomized controlled trials have shown a positive impact of interventions delivered by pharmacists in collaboration with physicians on glycemic control in outpatients with type 2 diabetes [9–11]. However, most clinical trials were carried out in developed countries and interventions and study design were poorly described [12]; this may hamper the implementation of the conducted clinical pharmacy services into others realities, the mapping the components of the pharmacist interventions and measure their actual effects. Besides, studies on pharmacist interventions have focused primarily on outcomes reporting and usually do not provide useful insights into factors that make an intervention successful in achieving outcomes [13].

This study aimed to (a) propose and evaluate the effectiveness of a pharmacist–physician collaborative care model on the reduction and control of A1C levels, blood pressure and low-density lipoprotein (LDL) cholesterol in outpatients with uncontrolled type 2 diabetes; and (b) determine the variables that could explain the reduction and individualized control of A1C.

Methods

This study was reported using the Consolidated Standards of Reporting Trials, and the pharmacist interventions were described based on the Descriptive Elements of Pharmacist Intervention Characterization Tool [13].

Design and setting

A randomized controlled trial of a pharmacist–physician collaborative model for patients with type 2 diabetes was conducted between September 2012 and April 2014. This study was conducted at a university hospital-affiliated secondary care clinic in São Paulo, Brazil. The metabolic disease clinic's medical staff consists of three specialist physicians who follow patients with diabetes, hypertension, or dyslipidemia who have been referred by the primary care physician.

Nurses and nutritionists also are part of the outpatient care team. Although the pharmacists are available in hospital, they meet the demand of inpatients and providing clinical services such as discharge counselling and prescription review. Outpatients were not receiving the clinical pharmacy services and, therefore, this study was designed and developed by one pharmacist-researcher who was not part of the clinical staff. This researcher previously received training on working process for clinical services (patient assessment, development of the care plan, evaluation of follow-up) and had experience in the care of patients with cardiovascular diseases.

The study protocol was approved and executed according to the requirements of the Ethics Committee at the University Hospital and Faculty of Pharmaceutical Sciences of University of São Paulo.

Patient selection and randomization

Potential participants were identified by pharmacist-researcher through medical record review before the regular consultation in diabetes clinic. If the patients met the study criteria, they were referred by the physician to pharmaceutical consultation. Male and female patients were eligible to participate if they were 40–79 years old; had been diagnosed with type 2 diabetes for at least 6 months, according to the Brazilian national consensus guidelines [14]; were using anti-diabetic drug; and had A1C levels ≥7% for no more than 4 months prior to the baseline of the study.

The exclusion criteria were pregnancy; end-stage renal failure (hemodialysis or glomerular filtration rate $< 10\,\mathrm{mL\,min^{-1}}$); dementia or unstable psychiatric conditions; liver disease; treatment for cancer, acquired immune deficiency syndrome or anticoagulation; plan to change residence during the study period; or participation in another study at the university hospital, assessing the impact of others health interventions on cardiovascular outcomes.

Eligible patients were recruited from September 2012 through April 2013, and were randomized in a 1:1 ratio to the intervention or control group. Randomization was performed by pharmacist-researcher using a computer-generated randomized list from the research randomizer programme (https://www.randomizer.org/), and followed the allocation sequence according to the referral of the physicians. Patients, physicians and pharmacist did not have foreknowledge of intervention assignments. After randomization, patients and the pharmacist were aware of the allocation of the control and intervention groups. The medical team was unaware of the patients who were randomized to the control group; but they knew on the intervention group, because any suggestions for adjustment of therapy were discussed with them.

All eligible patients were informed by pharmacist-researcher on the nature of the trial, and those who agreed to participate were asked to sign a written informed consent form, and both the pharmacist and the patients were left with a signed copy. From that moment, the participants were followed up for 12 months.

Description of usual care

Participants in the control group received only the usual care services provided by the metabolic disease clinic, which included regular follow-up clinic visits with the medical and nursing staff for 2–6 months (depending on glycemic control). During these visits, a physician assessed the patient's pharmacotherapy and clinical parameters, and, if it was deemed necessary, the patient was referred to the nursing team, which provided guidance on disease, lifestyle, self-monitoring of blood glucose and administration of insulin, as well as provided educational leaflets about delivery of diabetes care and a monthly self-monitoring diary.

Description of pharmacist-physician collaborative care model

In addition to the usual care, patients in the intervention group received a clinical pharmacist service. Because of the lack of reference on pharmacists care, we presented to the outpatient medical staff the type of service being offered, the clinical expertise of

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the pharmacist and how this could add to the patient care. After the establishment of this alliance for diabetes care, pharmacist and physician worked collaboratively on the same time and setting. The pharmacist conducted face-to-face individual consultations with the patient (lasting 20–40 minutes), after the routine secondary care visits. The patients also received 1 or 2 remote contacts by phone (lasting 3–20 minutes) during the interval between scheduled follow-up visits in the clinic.

Clinical data (co-morbidities, signs and symptoms and A1C, blood pressure, and LDL cholesterol), data on pharmacotherapy (prescription and over-the-counter drugs, patients' medication experience, medication adherence and medication accessibility), information on lifestyle (physical activity, diet, smoking and alcohol consumption) and sociodemographic data (age, gender, education and marital status) were collected from the patients or other sources available to the pharmacist (medical records, prescriptions, data on patients' self-monitoring of blood glucose, and laboratory test results). The analysis of such information guided the process of building the necessary pharmacist interventions.

During face-to-face consultations, health education-related issues were discussed (e.g. diabetes, lifestyle and self-monitoring of blood glucose), and specific information was provided about the use of medications (medication adherence; adverse reactions; storage; and administration, especially for insulin). The instructions were usually verbal and, depending on the needs of the patient, educational leaflets on diabetes care were delivered (e.g. about hypoglycemia), which are standardized and available to other care team members. A monthly self-monitoring diary standardized was provided to patients who had their own blood glucose monitor – commonly those using insulin, because the Brazilian Public Health System offers the device free.

The clinical pharmacist also conducted a comprehensive medication review to identify drug-therapy problems. Such problems were appraised using the methodology Pharmacotherapy Workup that evaluates aspects related to indication, effectiveness, safety and adherence [15]. The causes of these problems and their solutions were put to the medical team and/or patients and discussed. The pharmacist's suggestions to initiate, discontinue or adjust medications, and order laboratory tests were supported on evidence-based literature or guidelines [14,16]. At every visit, a medications list was prepared and updated, and the patient received a medication chart to assist in the correct use of their medicines.

During remote support by telephone, the pharmacist generally investigated the presence of possible side effects, reinforced the importance of medication adherence to achieve the individual goals of therapy, and checked if the dosage schedule, agreed to during face-to-face consultation, was convenient for the patient. If any detected problem required a change in the prescribed pharmacotherapy, a regular outpatient visit with the physician and the pharmacist was recommended.

Outcome measures

The primary outcomes of the study included the reduction and control of A1C levels and the percentage of patients with very poorly controlled diabetes (A1C level > 9%). Secondary outcomes of interest were reduction and control of blood pressure for patients with hypertension and reduction and control of LDL cholesterol for patients with dyslipidemia. The following therapeutic goals were established according to the clinical guidelines [16]: A1C level < 7% and individualized goals (A1C level < 8% for patients with a history of severe hypoglycemia, various co-morbid conditions, advanced complications, or difficult to attain glycemic control despite the use of multiple glucose-lowering agents, including insulin; and < 7% for other patients); blood pressure < 130/80 mmHg; and LDL cholesterol level < 100 mg dL^{-1}.

All laboratory tests were performed by the centralized laboratory of the hospital; and the frequency of laboratory tests were established based on the physician's clinical judgement of the patient's needs and in accordance with internal protocol. The nursing staff measured blood pressure before each scheduled outpatient visit, using an automatic blood pressure monitor (Microlife BP A100, Microlife, Shenzhen, China) and standard clinical practice. The outcome assessors were blinded and unaware to which group the patients had been assigned.

In addition, some process indicators were evaluated. Medication adherence was assessed using the 4-item Morisky–Green test [17]; it is a validated tool (alpha reliability = 0.61) that evaluates the patient's attitudes towards treatment and classifies as adherent patients if they answered negatively all 4 items on this test. The medication complexity was measured by the validated Medication Regimen Complexity Index, which considers the dosage form (Section A), dosing frequency (Section B) and additional directions (Section C) [18]. The number and consumption of anti-diabetic drugs were judged using data contained in the prescription. All process indicators were collected at baseline and at end of study.

Definition of a completed follow-up participation

Patients were lost to follow-up if they did not attend at least three appointments throughout the study or if it was not possible to obtain the final A1C level (after 12 ± 2 months of the start of the study). Patients who did not have LDL cholesterol test results at the beginning or end of the study were not excluded; the missing data have been identified along the manuscript.

Sample size

Sample size was calculated using a two-tailed test and indicated that to detect an absolute difference of 0.76% in A1C in favour of the intervention group [19] with $\alpha = 0.05$, $\beta = 0.1$ (90% power), and standard deviation of 1.0%, a sample size of 36 patients per group was required. An additional 10% was added to allow for patients dropping out of the study, making a target sample size of 80 patients (40 patients in the control group and 40 in the intervention group).

Data analysis

Statistical analyses were performed using R version 3.2.4-revised (RStudio, Boston, MA, USA) and included only data of the patients who completed this trial. Data were examined using Fisher Exact test for categorical variables. Two-sample comparisons were made using Student's *t*-tests for normally distributed variables and Mann-Whitney *U*-test or Wilcoxon signed-rank test for non-normally distributed data. The data were presented as frequencies (percentage), mean (standard deviation), median [IQR] or mean

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difference [95% CI]. For all analyses, a P value of < 0.05 was considered statistically significant.

In addition, multiple linear regression (model 1 = all patients of study; model 2 = intervention group) and multiple logistic regression (model 3 = all patients; model 4 = intervention group) models were used to determine the covariates that could explain the reduction and individualized control of A1C (<8% for patients with a history of severe hypoglycemia, various co-morbid conditions, advanced complications, or difficult to attain glycemic control despite the use of multiple glucose-lowering agents, including insulin; and < 7% for other patients), respectively.

The covariates were a priori specified and described according to the framework proposed by Pigott *et al.* [20] for complex intervention, as substantive features related to the target participants (age [< 60 years, ≥ 60 years for models 3 and 4]); marital status; gender; education; physical activity; baseline A1C level; diabetes diagnostic time (≤ 10 years, > 10 years for models 3 and 4) and to the interventions (change in the adherence score; change in the percentage of adherent patients (< 100%, 100% for models 3 and 4); change in the number of anti-diabetics drugs; change in medication complexity score; participation in the intervention group (models 1 and 3); and percentage of resolved drug-therapy problems (models 2 and 4).)

Each statistical model entered univariate way these variates (with a $P = 0.15$) in a stepwise multiple regression, retaining only those variables with a P value < 0.05 in the final model. The models were analysed using SAS ENTERPRISE GUIDE version 4.3 (SAS Institute Inc., Cary, NC, USA).

Results

Patient characteristics

Of the 80 patients recruited and randomized, 73 (37 in the control group and 36 in the intervention group; 91.3%) completed this study and were included in the comparative analysis (Fig. 1). Patients in intervention group performed a mean of 4.3 ± 0.7 face-to-face consultations with the physicians plus pharmacist and received a mean of 2.4 ± 0.8 telephone calls; the control group had a mean of 4.3 ± 0.8 visits to physicians. Baseline characteristics were similar between the two study groups (P > 0.05 for all comparisons) (Table 1).

Pharmacist interventions

A total of 189 drug-therapy problems were documented with a mean of 5.3 ± 2.4 per patient. All patients presented at least one problem. The problems identified were primarily related to drug safety ($n = 60$; 31.8%) and medication adherence ($n = 59$; 31.2%). Non-adherence was the problem more often related with patients' medication experiences, such as doubts about the need for the medications ($n = 23$; 12.2%) and not liking to take medications ($n = 10$; 5.3%).

Of all interventions to resolve these problems, 125 (66.1%) were negotiated and agreed upon directly between the patient and clinical pharmacist, especially about specific instructions on proper use of medication (e.g. take/use at a specific time/s and relation to food), and work to prioritize health goals, sharing how the medication can fit those goals (Table 2).

The physician was involved in the remaining interventions ($n = 64$; 33.9%), most habitually to change the dosage regimen ($n = 26$; 13.8%) and change the drug product ($n = 15$; 7.9%) (Table 2). The interventions were discussed face-to-face with the physician and almost all the suggestions made by the pharmacist were accepted (96.9%) by physician. Most drug-therapy problems ($n = 169$; 89.4%) were solved or prevented at the end of this trial.

Process indicators

The mean number of medications (6.9 vs. 7.6; $P < 0.001$) and directions in the prescriptions (6.2 vs. 6.6; $P = 0.041$) increased significantly in the intervention group, but this change was not higher compared with the control group. On the other hand, we noted a significant increase in average consumption of Neutral Protamine Hagedorn (NPH) insulin in the control group (37.4 vs. 40.0; $P = 0.014$). No significant changes between the intervention and control groups were observed regarding the number of drugs, the consumption of metformin and NPH insulin and the medication complexity. The intervention group showed a significant increase in the percentage of adherent patients (33.3%; $P = 0.009$) and the average score from the Morisky–Green test (0.9; $P < 0.001$). These improvements were significantly higher compared with the control group (33.3% vs. −5.4% [$P < 0.001$]; and 0.9 vs. 0.1 [$P = 0.008$], respectively) (Table 3).

Clinical outcomes

The intervention group had significant reductions in A1C (−0.79; $P < 0.001$), systolic and diastolic blood pressures (−12.5 mmHg [$P < 0.001$] and −6.3 mmHg [$P = 0.011$], respectively) and LDL cholesterol levels ($-10.2\ \mathrm{mg\,dL^{-1}}$; $P = 0.041$), whereas only the LDL cholesterol level was significantly reduced in the control group ($-12.2\ \mathrm{mg\,dL^{-1}}$; $P = 0.025$). Also, there was a significant increase in the percentage of patients in the intervention group who achieved the recommended goals for A1C (< 7%: 16.7% [$P = 0.025$], and individualized goal: 25.0% [$P = 0.002$]) and systolic and diastolic blood pressures (41.9% [$P = 0.002$] and 32.2% [$P = 0.033$], respectively). Compared with usual care, patients in the intervention group showed a greater reduction in A1C levels (−0.79 vs. −0.16; $P = 0.010$) and an increase of patients achieving the individualized goal of A1C (25.0% vs. 5.4%; $P = 0.020$) and systolic blood pressure (41.9% vs. 15.2%; $P = 0.026$) (Table 4).

Predictors of reduction and individualized control of A1C

The linear regression showed that mean differences in A1C levels increased with higher baseline A1C (regression coefficient = −0.33 [95% CI: −0.49, −0.18], $P < 0.001$ for model 1; −0.27 [95% CI: −0.50, −0.03], $P = 0.024$ for model 2). In addition, change in medication adherence score (regression coefficient = −0.41 [95% CI: −0.59, −0.23], $P < 0.001$) and change in percentage of adherent patients (regression coefficient = −0.89 [95% CI: −1.48, −0.28], $P = 0.005$) were predictors of reduction of A1C for models 1 and 2, respectively. For logistic regression, model 3 revealed that receiving pharmacist interventions increased about 6 times the

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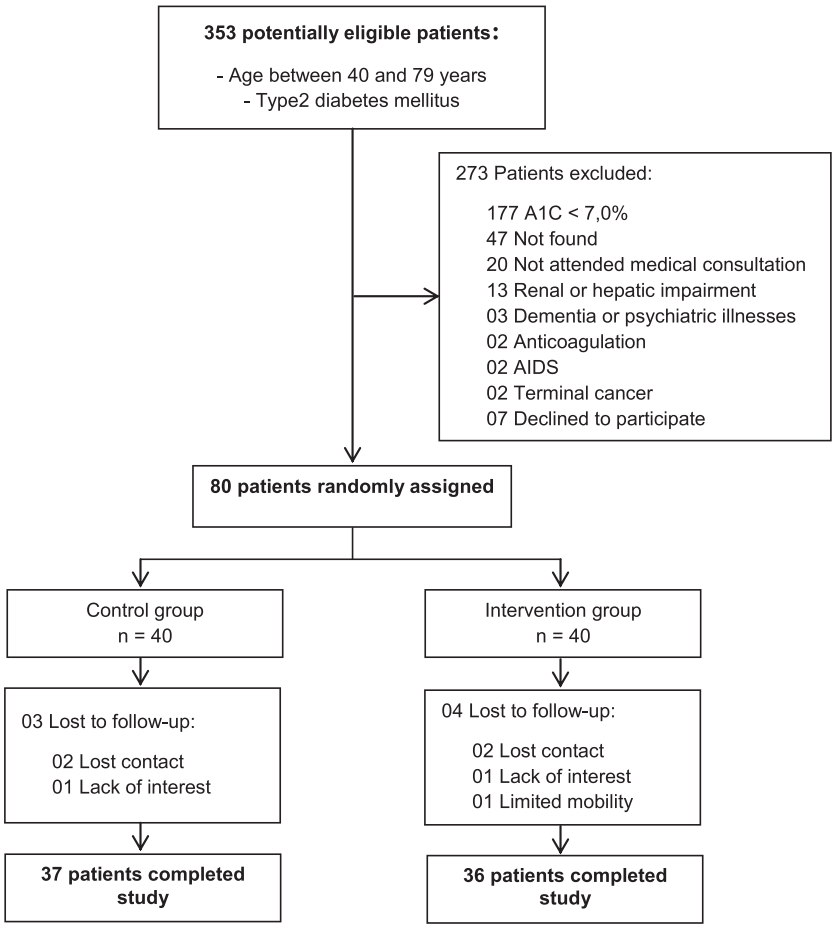

Figure 1 Flowchart of Study Patients. A1C, glycosylated haemoglobin; AIDS, acquired immune deficiency syndrome.

chance of achieving the individualized goal of A1C levels compared with the control group (odds ratio = 5.83 [95% CI: 1.16, 29.25], $P = 0.032$).

Discussion

The results indicate that the proposed physician–pharmacist collaborative model is effective for patients with uncontrolled type 2 diabetes, and they add to the literature on this theme. In Brazil, most of the evidence of the benefit of pharmacist interventions in diabetes care derives from studies conducted in community pharmacies / primary care, without a control group or nonrandomized study design [21–25]. Internationally, most randomized controlled trials have reported results of pharmacist interventions in non-diabetes outpatient clinics [12]. Also, our study provided insights on characteristics related to the examined population (higher baseline A1C) and intervention mechanism (higher change in medication adherence) that influence the effectiveness of pharmacist interventions.

In comparison with usual care, patients in the intervention group showed greater reduction in A1C levels. The mean difference detected between the groups was similar to the results of meta-analysis conducted by Machado *et al.* (−0.62%) [26], but lower than the finding of Collins *et al.* (−0.76%) [19] and Aguiar *et al.* (−0.85%) [12]. These meta-analyses included different practice settings, severity of disease and components of pharmacist interventions; then, we believe that there may be an effect size difference compared with our diabetes outpatient clinic. In addition, a multiple regression analysis indicated that our pharmacist-physician collaborative care model was more effective for patients with poor glycemic control, which is similar to findings of recent meta-analysis on pharmacist interventions for patients with diabetes [12].

The percentage of patients achieving glycemic control (A1C < 7%) increased in the intervention group, and this improvement tended to be higher compared with usual care group ($P = 0.056$). In similar trials in secondary care for patients with uncontrolled type 2 diabetes, the results were conflicting [27,28]. It is essential to remember that our study sample comprised patients with more difficulty to control the disease (e.g. old age, longer duration of diabetes, multiples co-morbidities, use of multiple anti-diabetic drugs, including insulin); these factors can lead to an increased risk for damage caused by the aggressive lowering of blood glucose. For that reason, less-stringent goals for A1C (i.e. < 8%) were commonly set for certain patients. If we consider the individual goals, the intervention group showed an increase in

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Table 1 Patient characteristics at baseline

Variable	Intervention ($n = 36$)	Control ($n = 37$)	*P* value
Female gender	25 (69.4)	24 (64.9)	0.804
Age (years)	61.1 ± 7.9	62.4 ± 8.2	0.504
Elderly (≥ 60 years)	21 (58.3)	20 (54.1)	0.815
Education until middle school	23 (63.9)	21 (56.8)	0.634
Married	22 (61.1)	25 (67.6)	0.630
Physical activity*	13 (36.1)	9 (24.3)	0.315
Diagnostic time (years)	12 [5.8 – 20]	10 [10 – 20]	0.794[§]
Co-morbidities			
Hypertension	31 (86.1)	33 (89.2)	0.736
Dyslipidemia	32 (88.9)	33 (89.2)	1.000
N° of medications	7 [5.8 – 9]	7 [6 – 8]	0.973[§]
MRCI	24 [16 – 27.3]	22 [18 – 24]	0.262[§]
N° of anti-diabetic drugs	3 [2 – 3]	3 [2 – 3]	0.692[§]
Biguanides	30 (83.3)	32 (86.5)	0.754
Insulins	27 (75.0)	28 (75.7)	1.000
Sulfonylureas	15 (41.6)	13 (35.2)	0.634
Gliptins	3 (8.3)	2 (5.4)	0.674
Consumption of anti-diabetic drugs			
Metformin (mg day^{-1})	1895.8 ± 1001.3	2117.6 ± 904.3	0.324
NPH Insulin (IU day^{-1})	44.5 [7.1 – 72.5]	36.0 [10 – 50]	0.434[§]
A1C (%)	9.03 ± 1.3	8.87 ± 1.4	0.613
A1C > 9%	17 (47.2)	15 (40.5)	0.640
SBP (mmHg)[†]	141.1 ± 16.7	142.8 ± 19.7	0.761
SBP <130 mmHg[†]	8 (25.8)	8 (24.2)	1.000
DBP (mmHg)[†]	78.7 ± 10.8	76.8 ± 11.6	0.495
DBP < 80 mmHg[†]	14 (45.2)	18 (54.5)	0.617
LDL cholesterol (mg dL^{-1})[‡]	93.0 [74.8 – 116.8]	98.5 [77 – 112.5]	0.847[§]
LDL cholesterol < 100 mg dL^{-1}[‡]	17 (60.7%)	12 (50.0%)	0.577
Medication adherence	14 (38.9)	16 (43.2)	0.813
Adherence score	3 [2 – 4]	3 [2 – 4]	0.237[§]

Data are defined as frequency (percentage), mean ± SD or median [IQR], as appropriate.
*At least 3 times a week, 30 minutes day^{-1}.
[†]Only for patients diagnosed with hypertension: intervention group ($n = 31$) and control group ($n = 33$).
[‡]Only for patients diagnosed with dyslipidemia and laboratory tests available before and after the study: intervention group ($n = 28$) and control group ($n = 24$).
[§]P values from Mann–Whitney *U*-test; other continuous variables were compared using parametric test.
A1C, glycosylated haemoglobin; DBP, diastolic blood pressure; IQR, interquartile range; MCRI, Medication Regimen Complexity Index; NPH, Neutral Protamine Hagedorn; SBP, systolic blood pressure; SD, standard deviation.

Table 2 Pharmacist interventions implemented directly by physician or patient to resolved drug-therapy problems

Actions required	*n* (%)
Patient-pharmacist	
Specific instructions on proper use of medication	61 (32.3%)
Patient-specific health education	57 (30.2%)
Drug administration device provided	4 (2.1%)
Discontinue the drug regimen not prescribed	2 (1.1%)
Physician-pharmacist	
Change the dosage regimen	26 (13.8%)
Change the drug product	15 (7.9%)
Initiate new drug therapy	13 (6.9%)
Removal of barriers to obtain medication	5 (2.6%)
Institute a monitoring plan (labs)	4 (2.1%)
Discontinue the drug regimen	2 (1.1%)

glycemic control rate higher than that seen in the control group. The practice of individualizing the A1C goals is consistent with the recommendation of the clinical guideline used as a reference in this study [16].

Along with better glycemic control, the intervention group showed significant improvement in reducing and controlling blood pressure and reducing LDL cholesterol levels. However, only the percentage of patients in the intervention group who achieved control of systolic blood pressure was statistically higher than that seen in the control group. A current meta-analysis [29] indicated that pharmacist interventions for patients with diabetes are associated with significant reductions in blood pressure and LDL cholesterol levels, as compared with the usual care. We believe this study probably does not present sample size and power sufficient to detect differences in the secondary clinical outcomes. Still, the reported results are particularly relevant considering that

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Table 3 Comparison of process indicators changes between intervention and control groups

Variable	Intervention ($n = 36$)		Control ($n = 37$)		*P* value between groups
	Change at 12 months	*P* value	Change at 12 months	*P* value	
N° of medications	0.6 [0.2, 0.9]	< 0.001*	0.1 [−0.3, 0.5]	0.609	0.078
MRCI total	0.7 [−0.3, 1.7]	0.229	1.0 [−0.1, 2.1]	0.070*	0.447
MRCI section A/dosage form	0.2 [−0.1, 0.5]	0.346	0.2 [−0.1, 0.5]	0.346	0.989
MRCI section B/dose frequency	0.1 [−0.6, 0.8]	0.814*	0.4 [−0.3, 1.1]	0.266	0.573*
MRCI section C/additional directions	0.4 [0.0, 0.7]	0.041	0.4 [−0.1, 0.8]	0.097	0.951*
N° of anti-diabetic drugs	0.14 [0.0, 0.3]	0.073	0.03 [−0.17, 0.23]	0.813	0.401
Metformin (mg day^{-1})	−87.5 [−297.3, 122.3]	0.959*	−79.7 [−298.6, 139.2]	0.833	0.695
Insulin NPH (IU day^{-1})	1.4 [−2.6, 5.4]	0.066	2.6 [0.0, 5.2]	0.014	0.945
Medication adherence	12 (33.3)	0.009	−2 (−5.4)	0.813	<0.001
Adherence score	0.9 [0.5, 1.4]	< 0.001	0.1 [−0.2, 0.5]	0.383	0.008

Data are expressed as frequency (percentage), or mean difference [95% CI], as appropriate.
**P* values from Student's t-tests; other continuous variables were compared using non-parametric tests.
MCRI, Medication Regimen Complexity Index.

Table 4 Comparison of clinical outcome changes between intervention and control groups

Variable	Intervention ($n = 36$)		Control ($n = 37$)		*P* value between groups
	Change at 12 months	*P* value	Change at 12 months	*P* value	
A1C (%)	−0.79 [−1.1, −0.5]	< 0.001	−0.16 [−0.5, 0.2]	0.632	0.010
A1C control (<7%)	6 (16.7)	0.025	1 (2.7)	1.000	0.056
A1C very poor controlled (>9%)	−8 (−22.2)	0.085	−3 (−8.1)	0.630	0.112
A1C individualized goal*	9 (25.0)	0.002	2 (5.4)	0.493	0.020
SBP (mmHg)†	−12.5 [−18.5, −6.5]	< 0.001§	−4.4 [−11.0, 2.3]	0.122	0.071§
SBP control (<130 mmHg)†	13 (41.9)	0.002	5 (15.2)	0.291	0.026
DBP (mmHg)†	−6.3 [−10.7, −1.9]	0.011	−3.2 [−7.4, 1.1]	0.142§	0.300§
DBP control (<80 mmHg)†	10 (32.2)	0.033	6 (18.2)	0.200	0.252
LDL cholesterol (mg dL^{-1})‡	−10.2 [−20.1, −0.3]	0.041	−12.2 [−21.2, −3.1]	0.025	0.768§
LDL cholesterol control (<100 mg dL^{-1})‡	1 (3.6)	1.000	5 (20.8)	0.238	0.084

Data are expressed as frequency (percentage), or mean difference [95% CI], as appropriate.
*A1C <8% for patients with history of severe hypoglycemia, advanced complications, extensive co-morbid conditions and difficult to attain glycemic control despite the use of multiple glucose lowering agents including insulin; and <7% for other patients.
†Only for patients diagnosed with hypertension: intervention group (n = 31) and control group (n = 33).
‡Only for patients diagnosed dyslipidemia and laboratory tests available before and after the study: intervention group (n = 28) and control group (n = 24).
§P values from Student's t-tests; other continuous variables were compared using non-parametric tests.
A1C, glycosylated haemoglobin; DBP, diastolic blood pressure; SBP, systolic blood pressure.

reductions in blood pressure and LDL cholesterol levels in patients with diabetes are strongly associated with cardiovascular risk reduction [30].

During the study, all patients presented at least one drug-therapy problems. Most problems involved non-adherence, which agrees findings from other studies conducted in Brazil [21,22]. Among the barriers identified, most patients reported doubts about the need for the medications and not liking to take medications. Our findings confirmed that the medication experience (e.g. values, desires and fears associated with treatment and care) could influence the development of non-adherence behaviour [31]. This barrier is differing to issues commonly described in the literature, such as not understanding the instructions or the difficulty for medication access [32]. In this sense, the understanding of the patient's medication experience was important to care process, possibly being a factor that mediated the improvement in medication adherence and, consequently, reach of higher reductions for A1C levels.

Most recommendations to medical staff involved the replacement or addition of a new medication or a dose adjustment to control type 2 diabetes or other co-morbidities unresponsive to conventional treatment. The physician accepted almost all of them, with the acceptance rate higher or similar to other trials in outpatient clinics [21,27]. Perhaps the success of the collaborative process was because both professionals work side by side (same time and setting), which facilitated discussions regarding adjustments in the pharmacotherapy. In addition, the medical team was aware of the study and agreed on the responsibility of the pharmacist in the care process of patients with type 2 diabetes. It is worth emphasizing that remains unclear if the

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pharmacist intervention will work the same way in other health systems or cultures if the contextual elements discussed here are not present.

Another interesting finding is that a significant improvement in A1C levels was achieved in the intervention group without significant change in the consumption of anti-diabetic drugs or the medication complexity. Given this context, we believed that a comprehensive medication review and medication counselling according to patient's needs contributed to the success of our findings, because the study sample had previously received health education on diabetes. In addition, this trial was patient-centred since their perspectives and needs were incorporated by the pharmacist during the clinical judgement and decision-making processes; and they were encouraged to discuss and reflect on problem situations related to health behaviour and use of medicines. According to Bolen *et al.* [33], the active involvement of patients with uncontrolled type 2 diabetes within the health-care setting could result in a strong cumulative impact on morbidity and mortality.

This present study has some limitations. First, it is essential to note that the high glycemic-control rate observed in the recruitment of patients contrasts with national and international findings [3,34]. Also, this study was conducted in a single location by one pharmacist, which limits the generalization of our findings. The sample-size calculation did not consider other key outcomes for the control of cardiovascular risk in patients with diabetes, such as blood pressure and lipid profile, but the target sample size of 40 patients per group was considered suitable and practical. Humanistic outcomes such as quality of life and patient satisfaction were not assessed in this study, given limited time available for the pharmaceutical consultation. Economic analyses were not performed, because not all data needed for the calculations involved were available, which is why the cost-effectiveness of the developed service model was not determined.

In addition, the physicians were not blinded to the clinical pharmacy service, which may have had some effect on the care of patients with diabetes in both study groups. Some of the patients in this study had no laboratory tests of LDL cholesterol levels. Finally, only data of the patients who completed this study were analysed, but the lost to follow-up was balanced between the groups and no difference was noted for the characteristics of patients and reasons for withdrawal. Despite the limitations discussed here, the offered study design was intended to minimize the influence of the pharmacist practice on the usual routine outpatient clinic and thereby try to reflect the real-world effectiveness of the physician–pharmacist collaborative care model.

In conclusion, this study show that the pharmacist-physician collaborative care model developed is feasible and more effective than usual care in the reduction and individualized control of A1C levels in patients with uncontrolled type 2 diabetes. This effect size was achieved without significantly change the consumption of anti-diabetic drugs or the medication complexity. In addition, participation in the intervention group, higher baseline A1C and greater change in medication adherence were all significant predictors of improvement in A1C levels. Future research is required to monitor the long-term benefits and the sustainability of the proposed care model.

Conflict of interest

The authors declare no conflict of interest.

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